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ಬೆಳಗಾವಿ, ಕರ್ನಾಟಕ

A REPORT ON

"Insurance Pricing Forecast"

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Abstract

Insurance pricing is a fundamental component of the insurance industry, where determining accurate premium amounts is essential for maintaining a balance between risk coverage and profitability. Traditional pricing methods often rely on statistical averages and manual actuarial calculations, which may not effectively capture complex, non-linear relationships between multiple risk factors. With the rapid growth of data and computational power, techniques from Machine Learning have emerged as powerful tools to improve the accuracy and efficiency of insurance pricing.

This project presents a machine learning-based system for forecasting insurance premiums by analysing historical data and identifying hidden patterns among various influencing factors. The dataset typically includes attributes such as age, gender, body mass index (BMI), smoking habits, number of dependents, medical history, and geographical location. These features play a significant role in determining the risk profile of an individual, which directly impacts the insurance cost.

The proposed system follows a structured pipeline that begins with data collection and preprocessing, including handling missing values, encoding categorical variables, and normalizing data for better model performance. Exploratory Data Analysis (EDA) is performed to understand feature distributions and correlations, helping in selecting the most relevant variables for prediction. Various regression-based machine learning models, such as Linear Regression, Decision Trees, and ensemble methods like Random Forest, are applied to train the system.

The trained models are evaluated using performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R^2) score to ensure accuracy and reliability. The best-performing model is then used to predict insurance premiums for new user inputs. This automated approach reduces human intervention, minimizes errors, and enhances decision-making processes.

The system provides several advantages, including improved prediction accuracy, scalability, and adaptability to new data. It enables insurance companies to design fair and personalized pricing strategies while allowing customers to gain transparency in how premiums are calculated. Additionally, the model can be extended and integrated into real-time applications, making it suitable for deployment in web-based or enterprise-level systems.

In conclusion, this project demonstrates the practical application of Machine Learning in solving real-world problems in the insurance domain. By leveraging data-driven techniques, the system not only enhances pricing efficiency but also contributes to innovation in financial and risk management services.

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Introduction

In the insurance industry, pricing policies accurately is essential to balance profitability and customer satisfaction. Traditional methods rely heavily on manual calculations and actuarial assumptions, which may not fully capture complex relationships in data.

Machine learning provides an advanced approach by learning patterns from large datasets and improving prediction accuracy. In this project, models are built using techniques from Machine Learning to forecast insurance premiums. Algorithms such as regression models are trained on historical datasets to predict costs based on multiple input features.

1.1 Preamble

In the modern financial ecosystem, insurance plays a crucial role in providing protection against uncertainties and risks. One of the most important aspects of the insurance industry is determining the appropriate premium for each individual policyholder. Insurance pricing is not just a financial calculation; it is a complex decision-making process that depends on multiple factors such as age, health condition, lifestyle habits, and economic conditions. Accurately estimating these premiums is essential to ensure fairness for customers while maintaining profitability for insurance companies.

Traditionally, insurance pricing has been carried out using actuarial methods based on historical averages and statistical assumptions. While these methods have been effective to some extent, they often fail to capture complex relationships among multiple variables, especially when dealing with large and diverse datasets. As the volume of data continues to grow, traditional approaches become less efficient and more prone to inaccuracies. This creates a need for more advanced, scalable, and intelligent systems that can analyze data effectively and generate precise predictions.

With advancements in technology, techniques from Machine Learning have emerged as a powerful solution to address these challenges. Machine learning enables systems to learn patterns from historical data and make predictions without being explicitly programmed for every scenario. It is particularly useful in insurance pricing, where multiple interacting factors influence the final premium. By using algorithms such as regression models and ensemble learning methods, machine learning can uncover hidden patterns and provide more accurate and data-driven predictions.

LITERATURE FINDINGS AND PROBLEM STATEMENT

2.1 Background

Insurance pricing has traditionally been handled using actuarial science, where experts analyze historical data and apply statistical models to estimate risk and determine premiums. These methods rely heavily on assumptions, aggregated data, and linear relationships between variables. While effective in earlier stages, they struggle to handle the complexity and scale of modern datasets.

With the growth of digital records, insurance companies now collect large volumes of structured and unstructured data, including demographic details, medical history, lifestyle habits, and behavioural patterns. Traditional statistical approaches are limited in capturing non-linear interactions among these variables.

2.2 LITERATURE FINDINGS

Several research studies and industry implementations highlight the effectiveness of machine learning in insurance pricing:

- Traditional models like linear regression have been widely used due to their simplicity and interpretability, but they often fail to capture complex feature interactions.
- Studies show that tree-based models such as Decision Trees and Random Forest provide better performance by handling non-linear relationships and feature importance effectively.
- Ensemble techniques (e.g., Gradient Boosting) have demonstrated higher accuracy in predicting insurance costs compared to single models.
- Research indicates that features such as smoking habits, age, and BMI have a strong influence on insurance premiums, making feature selection a critical step in model development.
- Some studies also highlight the importance of data preprocessing and handling bias, as poor-quality data can significantly impact model performance.
- Recent advancements combine Machine Learning with big data technologies to improve scalability and real-time pricing.

Overall, literature suggests that machine learning-based models outperform traditional actuarial methods in terms of accuracy, adaptability, and efficiency.

2.3 PROBLEM STATEMENT

Insurance companies face significant challenges in determining accurate and fair premium pricing due to the complexity and variability of risk factors involved. Traditional pricing methods rely on manual calculations and statistical assumptions, which are often insufficient to handle large volumes of data and fail to capture complex, non-linear relationships between variables such as age, BMI, lifestyle habits, and medical history. These limitations can lead to inaccurate pricing, lack of personalization, and increased chances of human error. Therefore, there is a need to develop an efficient and automated system that leverages techniques from Machine Learning to analyze historical data, identify hidden patterns, and predict insurance premiums accurately, ensuring improved decision-making, scalability, and fairness in pricing.

2.4 OBJECTIVES

The main objectives of developing an Insurance Pricing Forecast are:

1 To develop a machine learning model for predicting insurance premiums

2 To analyze the impact of various features (age, BMI, smoking habits, etc.)

3 To improve prediction accuracy using data-driven techniques

4 To automate the insurance pricing process

5 To gain practical experience in machine learning model development

2.5 Applications

- **Personalized Premium Calculation:** Calculates insurance cost based on individual risk factors
- **Risk Assessment:** Helps identify high-risk and low-risk customers
- **Automated Pricing:** Reduces manual effort and improves efficiency
- **Fraud Detection:** Identifies unusual or suspicious patterns
- **Customer Segmentation:** Groups users for better policy design
- **Real-Time Prediction:** Provides instant premium estimates in web apps
- **Policy Optimization:** Helps companies design better pricing strategies

2.6 Summary

Studies in the insurance domain show that traditional actuarial methods, while simple and widely used, are limited in handling large and complex datasets and often fail to capture non-linear relationships among multiple risk factors. Basic models such as linear regression are easy to interpret but lack the accuracy required for real-world applications.

In contrast, advanced models like decision trees, random forests, and gradient boosting have demonstrated significantly better performance by effectively identifying complex patterns in data. Research also highlights that feature selection, including factors such as age, BMI, and smoking habits, plays a crucial role in improving prediction accuracy. With the adoption of techniques from Machine Learning, insurance pricing has become more accurate, scalable, and automated. These advancements are widely applied in real-world scenarios such as health insurance premium calculation, vehicle insurance risk assessment, and life insurance pricing. Additionally, machine learning enables automated pricing systems, real-time premium prediction in digital platforms, and improved decision-making processes, making it a powerful tool in modern insurance applications.

Methodology

3.1 Flowchart

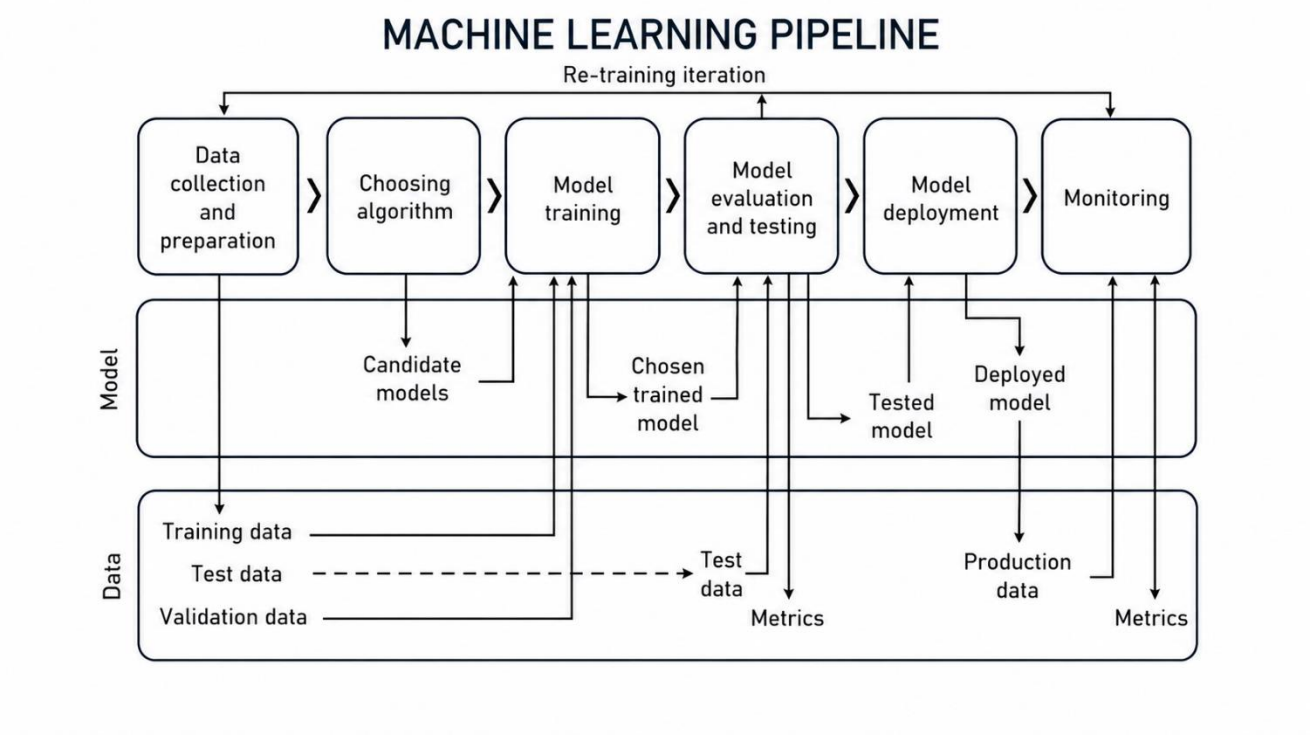

Fig 3.1 Flowchart of Insurance Pricing Forecast

1. Data Collection and Preparation

- This is the starting point of the pipeline.
- Data is collected from sources like datasets, APIs, or databases.
- Then it is **cleaned and prepared**:
    - Handling missing values
    - Removing duplicates
    - Encoding categorical data
    - Normalization/standardization

Output: **Clean dataset ready for modelling**

2. Choosing Algorithm

- Different algorithms are selected based on the problem type:
  - Regression (for price prediction)
  - Classification (for categories)
- Multiple models are tested (called **candidate models**)

Examples:

- Linear Regression
- Random Forest
- Gradient Boosting

3. Model Training

- The selected algorithms are trained using **training data**
- The model learns patterns between input (features) and output (target)

Important:

- Training data = data used to teach the model
- The model adjusts its internal parameters

Output: **Trained model**

4. Model Evaluation and Testing

- The trained model is tested using:
  - **Validation data** (for tuning)
  - **Test data** (for final evaluation)
- Performance is measured using metrics:
  - MAE (Mean Absolute Error)
  - MSE (Mean Squared Error)
  - R^2 Score

Output: **Tested model + performance metrics**

5. Model Deployment

- Once the model performs well, it is deployed into a real system
- It can be integrated into:
  - Web applications
  - Mobile apps
  - APIs

Now the model starts working on **real-world (production) data**

6. Monitoring

- After deployment, the model is continuously monitored:
  - Accuracy over time
  - Data changes
  - Errors or failures
- Real-world data changes → model performance may drop

7. Re-Training Iteration (Very Important)

- If performance decreases:
  - New data is collected
  - Model is retrained

This creates a **continuous learning loop**

3.2 CASE STUDIES

Case Study 1: Health Insurance Premium Prediction

A health insurance company implemented a machine learning model to predict premiums based on customer data such as age, BMI, smoking habits, and medical history. Initially, the company used traditional actuarial methods, which resulted in generalized pricing. After adopting models from Machine Learning like Random Forest, the prediction accuracy improved significantly. The study found that smoking status and BMI were the most influential factors. As a result, the company introduced personalized pricing, leading to better customer satisfaction and reduced financial risk.

Case Study 2: Vehicle Insurance Risk-Based Pricing

An automobile insurance provider used machine learning to calculate premiums based on driver behaviour, vehicle type, and accident history. By applying predictive models, the system could classify drivers into risk categories. High-risk drivers were charged higher premiums, while safe drivers received discounts. This approach reduced claim losses and improved profitability. The system also enabled real-time premium calculation in online platforms.

Case Study 3: Life Insurance Risk Assessment

A life insurance firm utilized predictive analytics to assess customer risk profiles using factors such as age, lifestyle, occupation, and health conditions. Machine learning models helped identify high-risk individuals more accurately than traditional methods. This improved underwriting decisions and minimized financial losses. The company also used the model to design better policy plans tailored to different customer segments.

Case Study 4: Fraud Detection in Insurance Claims

An insurance company integrated machine learning with its pricing system to detect fraudulent claims. The model analysed unusual patterns in claims data and flagged suspicious activities. This reduced fraudulent payouts and improved system reliability. It also helped in maintaining fair pricing for genuine customers.

Results

The insurance pricing forecast shows that premium cost is mainly influenced by factors like age, BMI, and smoking status. Smokers and individuals with higher BMI tend to have significantly higher predicted premiums. Age also plays a key role, with older customers generally paying more due to higher health risks.

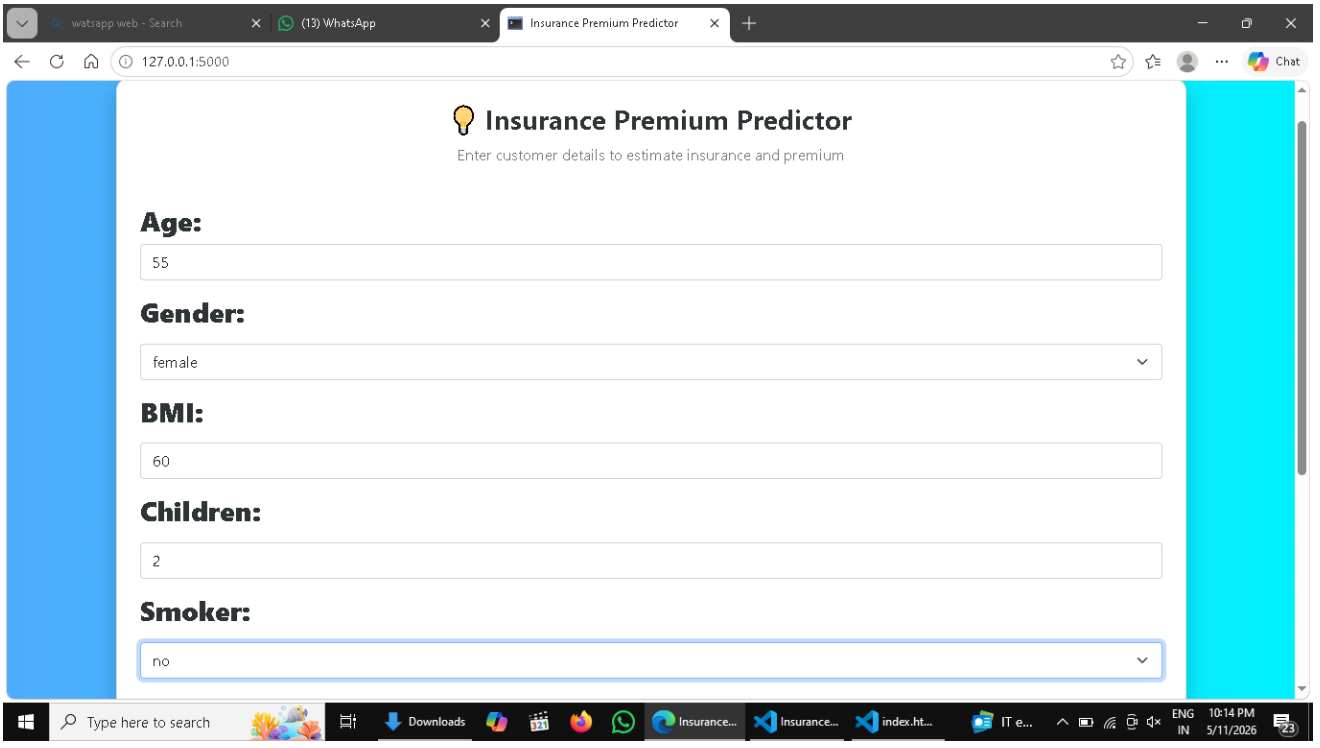

Fig 4.1 Web Interface

The above Fig 4.1 Shows web-based interface for an Insurance Premium Predictor application running locally (127.0.0.1). It collects user inputs such as age, weight, gender, BMI, number of children, smoking status, and region. These inputs act as features for a machine learning model to estimate the insurance premium. Dropdowns are used for categorical data like gender, smoker status, and region, while text fields capture numeric values. After entering all details, the user clicks the “Predict Premium” button to generate the result. The design suggests a simple front-end (likely HTML/CSS) connected to a backend model (commonly Flask or similar).

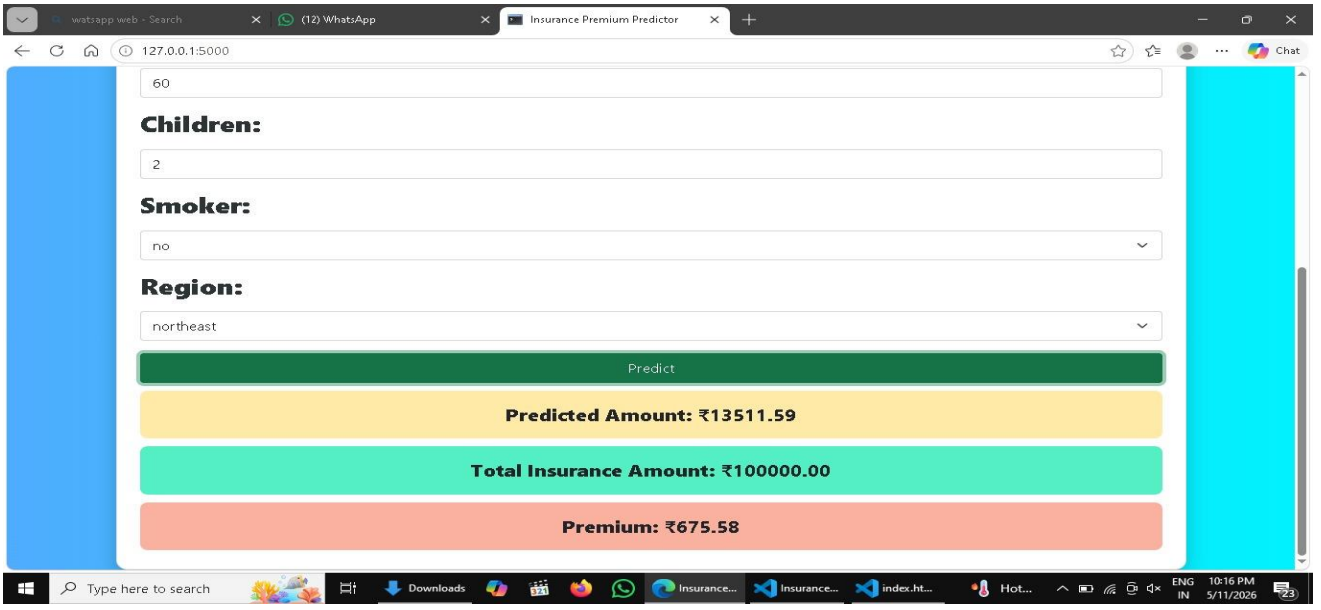

Fig 4.2 Prediction Result Page

The Fig 4.2 shows the prediction results generated by the Machine Learning model. The system displays the predicted insurance amount, total insurance coverage, and premium value. The prediction is based on the customer details entered in the form. This helps users estimate insurance costs quickly and accurately through an automated system.

Summary

The insurance pricing analysis indicates that health-related factors drive premium variation the most. Customers with risky profiles (like smokers or high BMI) fall into higher cost brackets. Most individuals are clustered in lower expense ranges, but a few high-cost cases increase overall variability. The model handles skewed data effectively and captures real-world cost patterns.

Conclusion

In conclusion, the Insurance Pricing Forecast using Machine Learning provides a robust and efficient solution to one of the most critical challenges in the insurance industry—accurate premium determination. This project demonstrates how traditional actuarial methods, which rely on assumptions and limited statistical models, can be significantly enhanced through data-driven techniques. By utilizing machine learning algorithms, the system is capable of analysing complex and non-linear relationships among various influencing factors such as age, BMI, smoking habits, and demographic details. The implementation process, which includes data collection, preprocessing, exploratory analysis, feature selection, model training, and evaluation, ensures that the developed model is both reliable and scalable.

The results clearly indicate that advanced models like Random Forest and Gradient Boosting outperform basic regression techniques in terms of accuracy and prediction capability. These models not only reduce prediction errors but also improve the overall efficiency of the pricing system. Additionally, the ability of the model to generalize well on unseen data confirms its suitability for real-world deployment. Another important outcome of this project is the identification of key factors influencing insurance costs, which enhances transparency and helps both insurers and customers better understand pricing mechanisms.

Furthermore, the system reduces manual effort, minimizes human errors, and supports automated decision-making processes. It also opens opportunities for real-time pricing, personalized insurance plans, and integration into digital platforms such as web and mobile applications. Despite certain limitations, such as dependency on data quality and the need for regular updates, the proposed approach proves to be highly effective and adaptable.

Overall, this project highlights the transformative impact of Machine Learning in the insurance domain by improving accuracy, efficiency, and fairness in pricing. It serves as a strong foundation for future advancements, including the use of deep learning models, real-time analytics, and large-scale deployment, ultimately contributing to smarter and more intelligent financial systems.

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ANNEXURES

Annexure 1

app.py

```
from flask import Flask, request, render_template
import pandas as pd
import pickle
app = Flask(_name_)
# Load model
with open("./gradient_boosting_regressor_model.pkl", 'rb') as file:
    model = pickle.load(file)
# Load dataset for dropdowns
data = pd.read_csv('./clean_data.csv')
@app.route('/')
def index():
    sex = sorted(data['sex'].unique())
    smoker = sorted(data['smoker'].unique())
    region = sorted(data['region'].unique())
    return render_template('index.html', sex=sex, smoker=smoker, region=region)
@app.route('/predict', methods=['POST'])
def predict():
    age = int(request.form.get('age'))
    sex = request.form.get('sex')
    bmi = float(request.form.get('bmi'))
    children = int(request.form.get('children'))
    smoker = request.form.get('smoker')
    region = request.form.get('region')
    # Model prediction
    prediction = model.predict(pd.DataFrame([[age, sex, bmi, children, smoker, region]],
                columns=['age', 'sex', 'bmi', 'children', 'smoker', 'region']))
    predicted_amount = prediction[0]  # raw predicted amount
    # Total insurance is always fixed at 1,00,000
    total_insurance = 100000
```

```
    # Premium calculation (5% of predicted amount, capped at total insurance)
    capped_amount = min(predicted_amount, total_insurance)
    premium_rate = 0.05
    premium = capped_amount * premium_rate
    # Return all three values separated by "|"
    return f"{predicted_amount}|{total_insurance}|{premium}"
if _name_ == "_main_":
    app.run(debug=True)
```

Annexure 2

Html file

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>Insurance Premium Predictor</title>
    <!-- Bootstrap CSS -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.1.0/dist/css/bootstrap.min.css"
rel="stylesheet">
    <style>
      body {
        background: linear-gradient(to right, #4facfe, #00f2fe);
        font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      }
      .card {
        margin-top: 40px;
        border-radius: 12px;
        box-shadow: 0 8px 20px rgba(0,0,0,0.2);
      }
      .result-box {
        border-radius: 8px;
```

```
      padding: 15px;
      margin-top: 10px;
      font-size: 1.3rem;
      font-weight: bold;
      text-align: center;
    }
    #predicted_amount { background-color: #ffeaa7; }
    #insurance_amount { background-color: #55efc4; }
    #premium_amount { background-color: #fab1a0; }
    h3 {
      font-weight: bold;
      color: #2d3436;
    }
  </style>
</head>
<body>
  <div class="container">
    <div class="card p-4">
      <h3 class="text-center">💡 Insurance Premium Predictor</h3>
      <p class="text-center text-muted">Enter customer details to estimate insurance and
premium</p>

      <form method="post" accept-charset="utf-8" class="mt-4">
        <div class="mb-3">
          <h3><label class center="form-label"><b>Age:</b></label><h3>
          <input class="form-control" id="age" name="age">
        </div>
        <div class="mb-3">
         <h3> <label class="form-label"><b>Gender:</b></label><h3>
          <select class="form-select" id="sex" name="sex">
            {% for type in sex %}
            <option value="{{ type }}">{{ type }}</option>
```

```
      {% endfor %}
    </select>
  </div>
  <div class="mb-3">
   <h3> <label class="form-label"><b>BMI:</b></label><h3>
    <input class="form-control" id="bmi" name="bmi">
  </div>
  <div class="mb-3">
    <h3><label class="form-label"><b>Children:</b></label><h3>
    <input class="form-control" id="children" name="children">
  </div>
  <div class="mb-3">
   <h3> <label class="form-label"><b>Smoker:</b></label><h3>
    <select class="form-select" id="smoker" name="smoker">
      {% for type in smoker %}
      <option value="{{ type }}">{{ type }}</option>
      {% endfor %}
    </select>
  </div>
  <div class="mb-3">
   <h3> <label class="form-label"><b>Region:</b></label><h3>
    <select class="form-select" id="region" name="region">
      {% for Region in region %}
      <option value="{{ Region }}">{{ Region }}</option>
      {% endfor %}
    </select>
  </div>
  <button class="btn btn-success w-100" onclick="send_data()">Predict</button>
</form>
<!-- Output boxes -->
<div id="predicted_amount" class="result-box">Predicted Amount: --</div>
<div id="insurance_amount" class="result-box">Total Insurance Amount: --</div>
```

```
      <div id="premium_amount" class="result-box">Premium: --</div>
    </div>
  </div>
  <script>
    function form_handler(event) {
      event.preventDefault();
    }
    function send_data() {
      document.querySelector('form').addEventListener('submit', form_handler);
      var fd = new FormData(document.querySelector('form'));
      var xhr = new XMLHttpRequest();
      xhr.open('post', '/predict', true);
      document.getElementById("predicted_amount").innerHTML = "Predicted Amount: ⏳";
      document.getElementById("insurance_amount").innerHTML = "Total Insurance Amount: ⏳";
      document.getElementById("premium_amount").innerHTML = "Premium: ⏳";
      xhr.onreadystatechange = function() {
        if (xhr.readyState == XMLHttpRequest.DONE) {
          // Response format: "predicted_amount|total_insurance|premium"
          let parts = xhr.responseText.split("|");
          let predicted_amount = parseFloat(parts[0]).toFixed(2);
          let total_insurance = parseFloat(parts[1]).toFixed(2);
          let premium = parseFloat(parts[2]).toFixed(2);
          document.getElementById("predicted_amount").innerHTML = "<b>Predicted Amount:</b> ₹" + predicted_amount;
          document.getElementById("insurance_amount").innerHTML = "<b>Total Insurance Amount:</b> ₹" + total_insurance;
          document.getElementById("premium_amount").innerHTML = "<b>Premium:</b> ₹" + premium;
        }
      }
      xhr.send(fd);
```

```
 }
   </script>
  </body>
</html>
```

Annexure 3

ML Training Model

```
Collecting pandas
  Using cached pandas-3.0.2-cp314-cp314-win_amd64.whl.metadata (19 kB)
Collecting numpy>=2.3.3 (from pandas)
  Using cached numpy-2.4.4-cp314-cp314-win_amd64.whl.metadata (6.6 kB)
Requirement already satisfied: python-dateutil>=2.8.2 in .\venv\Lib\site-packages (from pandas)
(2.9.0.post0)
Collecting tzdata (from pandas)
  Using cached tzdata-2026.1-py2.py3-none-any.whl.metadata (1.4 kB)
Requirement already satisfied: six>=1.5 in .\venv\Lib\site-packages (from python-dateutil>=2.8.2-
>pandas) (1.17.0)
Using cached pandas-3.0.2-cp314-cp314-win_amd64.whl (9.9 MB)
Using cached numpy-2.4.4-cp314-cp314-win_amd64.whl (12.4 MB)
Using cached tzdata-2026.1-py2.py3-none-any.whl (348 kB)
Installing collected packages: tzdata, numpy, pandas
```